Chapter 11
Using ChatGPT to Assist Writers with Planning Writing Tasks in an EFL Classroom

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11.1 Introduction

Although the release of ChatGPT in November 2022 was received with panic reactions among educational institutions and administrators, the field remains open to exploring its potential. Actors in both the industry and the educational sector advocate for the thoughtful utilization of ChatGPT, urging us to reevaluate curricula and contemplate the need for an educational revolution (Cope and Kalantzis 2024). These developments precede ChatGPT and refer to a wide range of anticipated effects of AI in education, including the alleviated burden on teachers towards the deliverance of quality educational materials (Luckin et al. 2016). In the previous decade, researchers have already focused on development (classification, matching, recommendation, and deep learning), application (feedback, reasoning, and adaptive learning), and integration of AI (affection computing, role-playing, immersive learning, and gamification). These early findings provide a promising channel for both learning and instruction while highlighting the changing roles of teachers and students in this emerging educational landscape of AI hegemony (Zhai et al. 2021).

Much of this research—and discomfort—pertaining to AI focuses on its interactive capacity, especially as it impacts assessment. One of its major effects is that, without any tools available to monitor and identify AI usage, teachers need to rethink take-home assignments that an advanced chatbot can resolve in seconds. Instead, instructors are called to emphasize formative assessment and design assignments to encourage critical thinking and problem-solving abilities. (Ali et al. 2023; Baidoo-Anu and Ansah 2023; Cope et al. 2020; Hong 2023; Mhlanga 2023). It is in this area that proponents of AI find the weight of its promise. Critical thinking re-emerges in importance and quality since it is essential in evaluating the responses

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generated by Generative AI (GAI), making an informed decision in using the information collected by it (Kasneci et al. 2023; Xiao and Zhi 2023), and requesting generations from it.

These elements of AI gain additional significance in language teaching and learning. The effectiveness of integrating ChatGPT in this area of educational praxis remains equally controversial, as research appears inconclusive (e.g., Basic et al. 2023) while remaining generally open to the possibilities: Hong (2023), for example, argues that ChatGPT can provide authentic language usage and serve as a personal tutor. In a similar vein, Kohnke et al. (2023) highlight its benefits in providing linguistic input, facilitating conversation practice, offering real-time assistance, and adjusting to learners' proficiency levels. AlAfnan's et al. (2023) empirical study investigated the usage of ChatGPT as an educational and learning tool for students and instructors of communication, business writing, and composition courses. Their results, too, incentivize exploring the potential of chatbots, such as ChatGPT, as helpful learning tools for collecting information and resources. Possibilities are soaring, but still, the mechanics of how it works remain relatively unexplored, especially in the intersections between AI, writing, and critical and metacognitive thinking.

This chapter addresses this intersection. It takes on, to that end, an intervention aiming to utilize ChatGPT as an aid to planning writing tasks in an EFL classroom in Taiwan, focusing on decision-making during interaction with the tool and metacognitive processing. Rather than prohibiting students from using ChatGPT when they compose, we explored the possibility of0 enhancing students' critical thinking and metacognitive awareness, enabling them to leverage ChatGPT for effective planning in their writing tasks. These attributes of human cognitive thinking, problem-solving abilities, and self-monitoring are identified by Li Jiang, AIRE Director at Stanford University (EO@Entreprenuership_Opportunities 2023), founder of CODE.org. Hadi Partovi (WSJ 2023) and other researchers (Kasneci et al. 2023) as crucial factors that distinguish human capabilities from those of machines. The question we are exploring, then, is an observed learning shift that asks learners to harness their cognitive abilities in order to maximize the advantages of AI technologies like ChatGPT.

The chapter is written in four parts. Our literature review explores scholarship related to human cognition, especially its relation between cognitive awareness, critical thinking, and writing. Addressing these interrelated terms means introducing the centrality of monitoring the cognitive process, especially as it translates into writing. The surveying of the literature alongside available models explaining the writing process serves to build a theoretical framework that informs the method and intervention we present in this chapter. The description of our intervention to a classroom of 26 students of Advanced English writing at a national university in Central Taiwan.

In this second part of this chapter, we detail the intervention, which aimed to teach students how to creatively and effectively utilize ChatGPT for writing tasks without compromising on integrity and quality. This mixed-methods research involving qualitative analysis of 257 prompts generated by students and their

ChatGPT output, as well as the quantitative analysis of questionnaires monitoring students' intentions, experiences, and impressions, lasted between February to June 2023. Our detailed analysis of their feedback and interactions with ChatGPT in the Discussion section of the chapter engages with students' motivations, experiences, and limitations and, as such, informs suggestions for educators and instructors on leveraging ChatGPT to support EFL learners. In return, rather than banning it in composition classrooms, we propose a critical utilization of ChatGPT in learners' writing tasks.

11.2 Metacognition and Writing: Definitions and Models

11.2.1 Metacognitive Awareness and Critical Thinking

While promoting the application of ChatGPT in teaching and learning, Kasneci et al. (2023) emphasize critical thinking and strategies as a requirement for educators and learners to take full advantage of large language models in education settings. In this chapter, as we propose a viable model for utilizing ChatGPT as a planning tool in writing classrooms, we build on their work, emphasizing the role of human cognition in the writing process and its relationship to critical thinking. We further explore how planning writing tasks with the help of ChatGPT encompasses *metacognitive awareness* and *critical thinking*.

We define metacognitive awareness in writing as an individual's ability to monitor their cognitive processes and activities during composition tasks consciously. Metacognitive awareness enables, thus, critical thinking to guide the writer in making informed decisions regarding planning for a writing task, considering factors that influence one's writing plans, such as task demands, rhetorical purpose, genre, and more. Metacognitive awareness plays a crucial role in the cognitive process by enabling learners to plan, monitor, evaluate, and regulate their cognitive activities throughout the task. Metacognitive awareness and metacognitive knowledge are essential in fostering students' abilities to be independent learners (Flavell 1979; Schraw and Moshman 1995; Veenman et al. 2006).

Critical thinking is a skill that enables individuals to analyze, evaluate, and synthesize information effectively. Paul and Elder (2006) define critical thinking as the intellectually disciplined process of actively and skillfully conceptualizing, applying, analyzing, synthesizing, and evaluating information gathered from observation, experience, reflection, reasoning, or communication. Halpern (1998, 2013) describes critical thinking as purposeful, reasoned, and goal-directed thinking that involves problem-solving, formulating inferences, calculating likelihoods, and making decisions. Learning empowers students with the abilities and strategies to gather, evaluate, organize, and present information while reflecting on their own ideas. Various instructional approaches, including individual study, dialogue, authentic or anchored instruction, and mentoring, are introduced in teaching critical thinking. These skills are essential for academic success and effective written communication.

More recent developments have turned to the 'metacognitive' capacities of machines, highlighting how AI already carries a large metacognitive weight, which may possibly translate to its pedagogical applications. Kasneci et al. (2023) and Ganapini et al. (2022) propose an architecture (SOFAI) inspired by human decision-making. It is supported by a meta-cognitive agent that assesses resource constraints and rewards to decide between them. The authors highlight the importance of metacognition in improving AI's decision-making abilities, and the proposed architecture aims to enhance AI's flexibility, performance, and decision quality. Moreover, a study by Drigas et al. (2023) introduces a nine-layered meta-learning pyramid model based on metacognition and technologies with intelligent traits. The model aligns with established theoretical models, emphasizing attention, memory, and metacognitive processes in regulating learning. The endeavors to incorporate the concept of metacognition into machine learning underscore the links between the metacognition of humans and that of machines, an aspect we assert can inform learners' interactions with ChatGPT.

11.2.2 Cognitive Activity of Writing: The Role of Metacognitive Monitoring

The questions that metacognitive awareness and critical thinking raise for educational praxis are inherent in writing. Planning as a cognitive process, involves a recursive cycle of setting goals, generating ideas, and organizing them (e.g., Flower and Hayes 1981; Grabe and Kaplan 2014; Hyland 2019); during the planning process, writers may undergo a repetitive cycle of the three cognitive activities. Writers can form writing plans or set writing goals based on several factors: task demands, available resources, time limits, etc. Writing goals can be divided into subgoals and may be revised during the writing process. Goals and subgoals formed by writers may be involved in cycles of generating and organizing ideas and engaging with critical thinking (Kuo and Anderson 2008). Two models of cognition in writing tasks impacted our research questions and study design.

The first is the framework of Flower and Hayes (1981) (Fig. 11.1). In line with their work, our study divides the planning stage into three subprocesses: generating, organizing, and goal setting. Furthermore, we engage with the concept of a monitoring mechanism as presented in their model. The monitoring mechanism oversees the entire cognitive process of composition.

In addition to Flower and Hayes' model, Grabe and Kaplan's (2014) writing model (Fig. 11.2) incorporates a metacognitive monitoring mechanism. While Flower and Hayes (1981) employed the concept of 'monitoring' without further elaborating on how it functions during the writing process, Grabe and Kaplan's (2014) work reflect this area with the relevant term of 'metacognitive processing' to

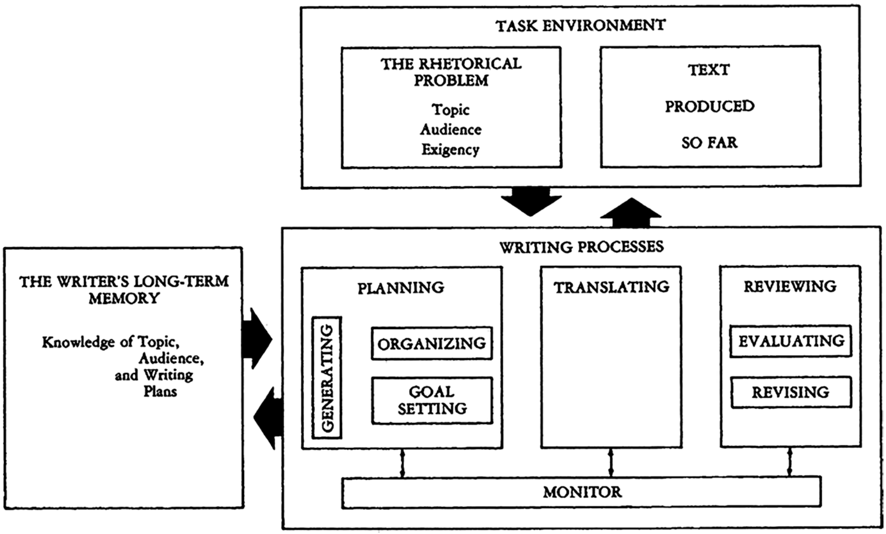

Fig. 11.1 Flower and Hayes' (1981) cognitive model of writing

explain how one's metacognitive awareness monitors his cognitive process of verbal composition and the resulting outcomes.

11.3 Methods

11.3.1 Research Questions

Drawing on an understanding of metacognition as monitoring, we studied writers' prompts to engage with ChatGPT. Our goal was to propose a model for instructors and learners to follow when they intend to use ChatGPT in planning writing tasks. To explore how EFL writers interact with ChatGPT and how ChatGPT can assist learners in planning English writing tasks, the following questions are addressed:

1. What prompts do EFL writers formulate during the planning phase of composing English writing tasks when interacting with ChatGPT?
2. How do EFL writers perceive the utilization of ChatGPT as a tool to aid in planning English writing tasks?
3. What factors influence the decision-making process of EFL writers in the generation of prompts?
4. What model can instructors and learners employ when integrating ChatGPT to assist writers in the planning phase of writing tasks?'

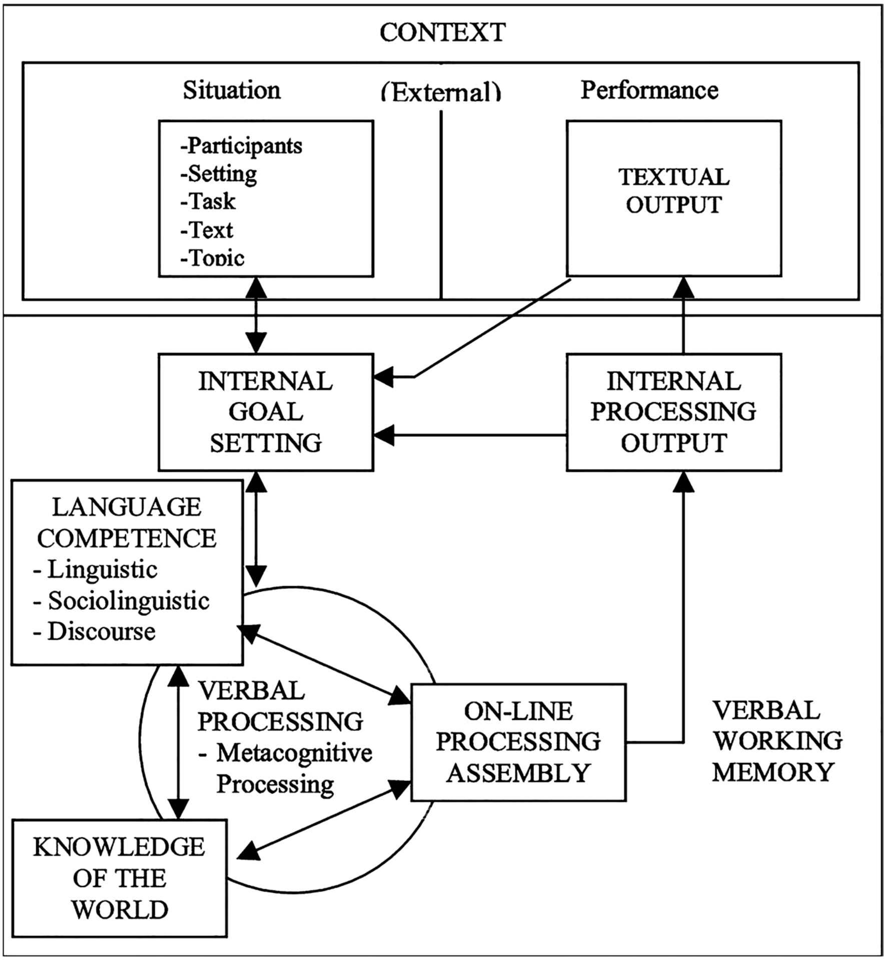

Fig. 11.2 Grabe and Kaplan's (2014) writing model

11.3.2 *Participants*

This study adopted a convenient sampling method to recruit English-major juniors from a national university located in central Taiwan as participants. A total of 26 student participants participated in the study. The selected students were enrolled in the required writing course, Advanced English Writing. The course contents include writing knowledge encompassing rhetorical structures of paragraphs and essays, style and genre formats and conventions, and writing mechanics. Course prerequisites included Basic English Writing and Intermediate English Writing courses during their freshman and sophomore years, as mandated by the department's curriculum. Students' native language is Mandarin Chinese, and they have been studying English since the third grade in elementary schools following the national curriculum in Taiwan.

Students were asked about their experiences using ChatGPT in the first period of the course. Only 3 students reported that they had any experience with ChatGPT before the class. In terms of digital literacy, they all had the experience of using word processors to complete assignments and make class presentations, and they reported that they used social media, such as Line, Instagram, Facebook, etc., in their daily lives.

11.3.3 Instruments

Questionnaire

The questionnaire was administered at the end of the semester; it was designed to understand the participants' perceptions of their engagement with ChatGPT. The questionnaire consisted of three subcategories: function, engagement, and application. The items in the function category inquired on the writing goals for which the participants employed ChatGPT. The engagement category asked questions regarding the participants' interactions with ChatGPT. The application category sought information on how research participants evaluated and monitored their interactions with ChatGPT.

Writing Tasks

Two writing tasks were assigned, before and after the intervention. The first writing task, which was assigned at the beginning of the semester when the students started to use ChatGPT in planning their English composition, was on aromatherapy, as shown below:

> The effects of aromatherapy remain controversial. Some believe that aromatherapy is beneficial for our health, while others claim that it can do harm to our body. What are your viewpoints toward aromatherapy? Compose an essay presenting your viewpoints on aromatherapy; you should take a definite position. Your essay should consist of an introduction, a minimum of two body paragraphs to support your arguments, and a concluding paragraph.

The second writing assignment was taught after the instructor explicitly taught the notion of a semantic map and demonstrated how to evaluate the responses generated by ChatGPT. It was on the 168 Intermittent Fasting Method, as shown below:

> The 168 Intermittent Fasting weight loss measure is getting popular recently. Some people think it is a feasible method of losing weight, while others hold doubts about it. What are your viewpoints toward the 168 Intermittent Fasting weight loss measure? Elaborate your arguments and use evidence and/or examples to support them.

Both writing tasks belonged to the genre of expository writing, and the task demands were clearly stated in the descriptions. The two topics were chosen for two specific reasons. Firstly, they are participants of controversy; diverse viewpoints exist regarding their effects. Secondly, while the students possessed some familiarity with these topics, their knowledge might not have been substantial enough to form compelling arguments supporting their stances. Consequently, the participants tended to rely less on their background knowledge to compose the assigned writing task; they instead needed to gather relevant information from sources such as the Internet. Thus, this motivated them to engage with ChatGPT during the planning phase of the writing task.

ChatGPT Prompts

The prompts used by the participants to interact with ChatGPT during both writing tasks were collected. The students were required to physically attend the designated classroom, bringing their computers to work on. ChatGPT was introduced to the students for the first time during the second instructional session. They were provided with the descriptions for the first writing task and encouraged to engage with ChatGPT to plan their assigned writing tasks. Each student was given 50 minutes to interact with ChatGPT on their computers. When the time was up, they were instructed to copy their interactions with ChatGPT and transfer the records into individual Google documents, which they shared with the instructor.

During the 14th instructional session, the students were required to compose the second writing task in the designated classroom. This time, they were given 100 minutes to engage with ChatGPT and finish their draft for the assignment. As they did in the first assigned writing task, they were instructed to copy their complete interactions with ChatGPT and transfer the records into individual Google documents, which were then shared with the instructor.

Throughout both composition tasks and their completion, students were working on their own personal computers, which granted them access to other search engines, such as Google, as they engaged with ChatGPT.

11.3.4 Instructional Guidelines

The intervention was designed based on the related literature surveyed in the previous sections and on the prompts the participants used to interact with ChatGPT during the pretreatment writing. The techniques introduced are based on the concepts of semantic mapping, critical thinking, and metacognition. Moreover, the course implemented explicit instruction to assist the EFL participants in effectively utilizing ChatGPT in the planning phase when they composed English compositions, which can enhance learners' metacognitive awareness and monitoring. The emphasis was on cultivating an understanding of the semantic map concept and developing critical thinking skills, enabling students to formulate prompts and use their critical thinking to make informed decisions during their interactions with ChatGPT.

Explicit Instruction

In this writing course, metacognition is integrated in two ways: explicit instruction and enhancing metacognitive monitoring. Explicit instruction is an approach that is often employed to improve learners' metacognitive awareness. It refers to a teaching approach or method in which educators provide clear and direct instruction to learners. It involves explicitly teaching specific concepts, knowledge, and strategies. In explicit instruction, teachers explicitly state the learning objectives, provide step-by-step explanations, model desired behaviors or thought processes, and offer guided practice opportunities. For instance, when instructing students on the strategy of clustering for idea generation during the planning phase, a teacher should explain what the strategy is, how the strategy is implemented, and why and when the strategy should be employed.

Thus, in this study, when introducing the concept of a semantic map, the instructor explicitly explained the definition of a semantic map, the process of creating one, its application in planning a composition, and the rationale for utilizing a semantic map, including when and why it is beneficial. By explicitly instructing students on implementing strategies, teachers can help them become more aware of their thinking processes, guiding them to think about their thinking. This can enhance students' metacognitive awareness: being consciously aware of their strengths and weaknesses and their cognitive processes and activities, students can make informed decisions and strategically adopt proper strategies to perform the tasks they can complete effectively.

Metacognitive Monitoring and Critical Thinking

To cultivate learners' metacognitive awareness and enable them to monitor their cognitive activities consciously, we instructed the participants to monitor and evaluate their engagement process with ChatGPT by employing self-questioning techniques. The questions are divided into two parts: the first required the students to think over the writing task, their writing activities, and processes; the second assisted them in evaluating their engagement with ChatGPT. Sample questions during the first part were exploratory on the knowledge of the participant themselves:

1. What are the task demands? What does the writing task require me to do?
2. What do I know about the topic? What else do I need to know to complete the task?
3. What stance should I take? What information do I need to form my arguments and support them?

Questions during the second part included:

1. What do I want ChatGPT to do?
2. Are the responses generated by ChatGPT relevant to my questions? If not, how can I rephrase them?
3. What further questions can I ask ChatGPT based on the previous responses?
4. Do I have enough information through the interactions with ChatGPT? If not, what are other possible sources of information?

Self-questioning techniques serve as valuable tools to enhance learners' conscious awareness of their cognitive process and activities when they undertake cognitive tasks, thereby cultivating their metacognitive awareness. By employing self-questioning, learners can actively monitor and evaluate their cognitive activities, enabling them to make informed judgments and effective decisions regarding cognitive tasks. We assume that with metacognitive awareness, the EFL participants could use their critical thinking to make informed decisions when they review the responses generated by ChatGPT and generate well-judged prompts based on previous interactions with ChatGPT.

Semantic Map

The concept of the semantic map was included in the instruction after prompts of the participants' first interaction with ChatGPT were analyzed. We observed that the prompts the participants formulated during their first interaction with ChatGPT appeared somewhat disorganized or 'scattered,' indicating a potential difficulty in organizing their ideas. Though free writing is one approach an individual may adopt when generating ideas for a writing task, we decided to integrate the concept of a semantic map, offering them an effective tool to organize both their ideas and the information generated by ChatGPT.

A semantic map is a visual representation or diagram illustrating the relationships between different concepts or ideas. It is often utilized to generate ideas. It serves as a tool to organize and structure the generated ideas to give users a clearer understanding of how various concepts are interconnected using visual representation. Figure 11.3 is an example of a semantic map, which the students drew manually on paper during the intervention. Using the structure of a semantic map, the students generated ideas related to the topic of living in a big city.

In the context of writing, the application of a semantic map involves utilizing this graphic representation to assist a writer plan for a writing task. It helps writers not only to brainstorm ideas but to organize them by identifying key themes and supporting details and by establishing relationships between different concepts. Its visual representation helps users map out the semantic connections between different concepts and the hierarchical relationships between them. Namely, in addition to generating ideas, semantic maps can further assist writers in organizing them to form a logical flow of information and thus enhance the coherence of their written work.

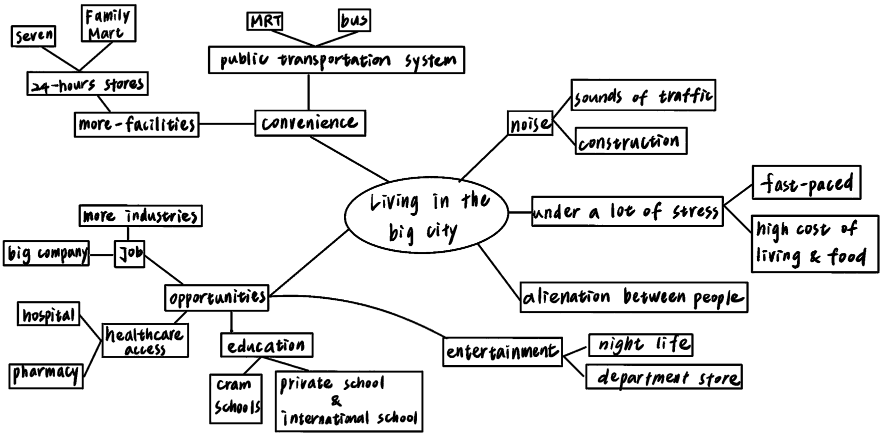

Fig. 11.3 A sample of semantic map on the topic 'Living in a big city' produced by a student participant

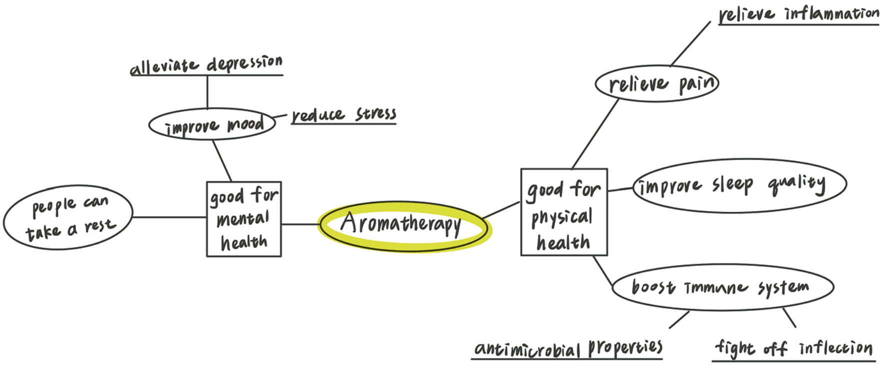

Fig. 11.4 A sample of semantic map on the topic 'aromatherapy' produced by a participant student

Following the principles of explicit instruction, the instructor taught the concept of semantic mapping. In addition to the teacher's demonstration, the students themselves applied the concept of semantic mapping in their planning process. For instance, the semantic maps shown in Figs. 11.3 and 11.4 were generated by the same participant. The semantic map in Fig. 11.3 was produced before the instructor provided explicit instruction on the concept of semantic map and its functions in the planning phase of a writing task, while the map in Fig. 11.4 was produced after receiving the instructor's instructions.

11.3.5 Data Collection and Data Analysis

The data collection procedures are summarized in Fig. 11.5.

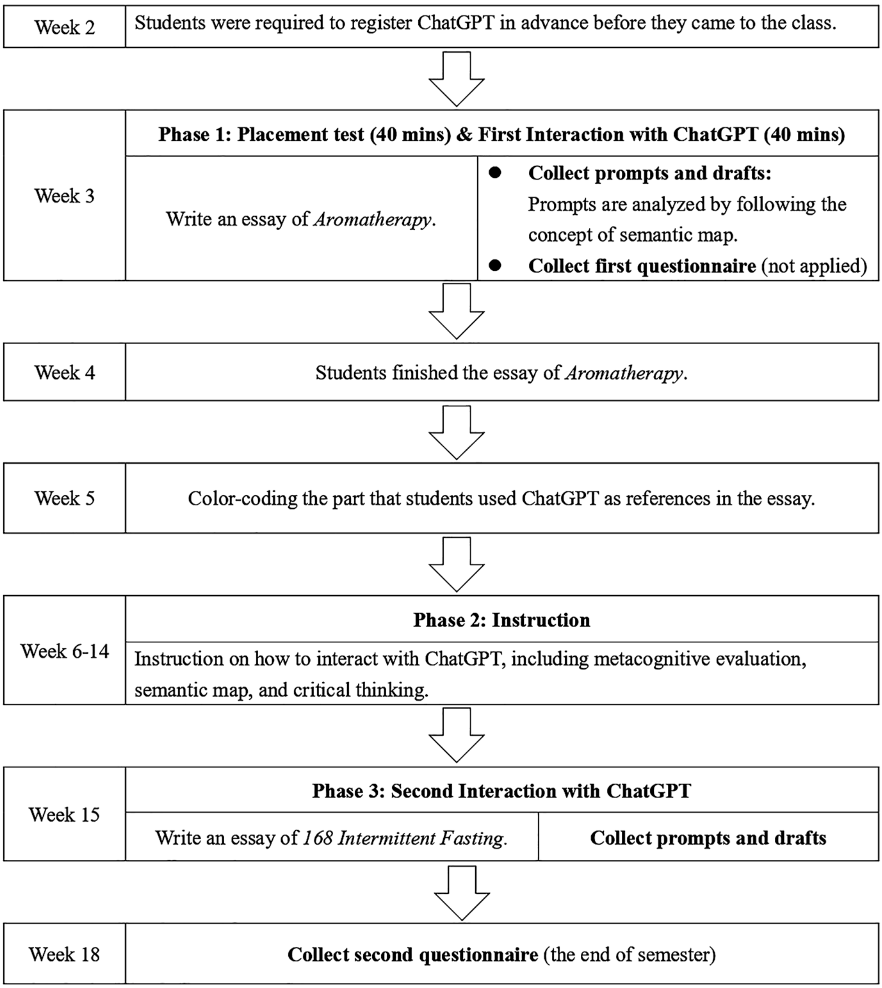

Fig. 11.5 Timeline of research. © 2024 Yu-ling You, Axson Lin, Shu-Jhen Han, Chiao-Chu Chu, and Chih-Yun Dai

A total of 257 prompts that the students composed to interact with ChatGPT were gathered and categorized by their primary purpose. These primary purposes were categorized into six types: concept, standpoint, argument, supporting detail, rhetoric sample, and evidence, as explained in Table 11.1. The provided examples, which were extracted directly from the collected data, were incomplete or ungrammatical sentences. Some prompts may not immediately appear to fit into specific categories; however, their functions were assessed based on the surrounding context. This involved considering the preceding and succeeding prompts, as well as the entire interactions with ChatGPT.

Table 11.1 The categories of prompts

Category	Definition	Examples
Concept	This category includes prompts where students ask ChatGPT to clarify or explain specific concepts, terms, or ideas related to their essay topic. Students ask these questions to assist themselves in having a clear understanding of key concepts before including them in their writing.	What is 168 intermittent fasting?
		What is the origin of using aromatherapy?
		What is skeptical?
		The history of 168 intermittent fasting.
Standpoint	In this category, students ask ChatGPT for guidance or suggestions regarding different perspectives or viewpoints they can consider in their essays. The prompts are used to obtain various angles and help students to make decisions on the standpoint of the essays.	Do you think 168 intermittent fasting is a good way to lose weight?
		Is aromatherapy effective?
		Is it scientific?
Argument	This category includes prompts where students seek assistance from ChatGPT in formulating their main argument or thesis statement for their essay. These prompts are adopted to assist students develop claims in their writing.	Benefits of 168 intermittent fasting.
		The danger of applying 168 intermittent fasting.
		What should we notice about 168 Intermittent Fasting?
Supporting detail	The prompts in this category are asked to require ChatGPT to provide more detailed information or examples that can serve as supporting details to bolster the arguments in students' own written texts. They are intended to assist students in strengthening their essays with relevant and persuasive information.	Why is this method easy to follow?
		Why can 168 intermittent fasting increase insulin sensitivity?
		Why are some people skeptical of Aromatherapy?
		How does 168 intermittent fasting help improve our heart health and brain health?

(continued)

Table 11.1 (continued)

Category	Definition	Examples
Rhetoric sample	The prompts in this category require ChatGPT to provide a sample outline or a model paragraph/essay on a specific topic. The responses by ChatGPT can help students observe and learn from well-structured language samples and coherent paragraphs to enhance their writing skills.	Give me the structure of an article on why people should not do Aromatherapy.
		Can you conclude it into a five-paragraph article with an introduction, three-paragraph body, and a conclusion?
Evidence	The prompts in this category are employed to request ChatGPT for specific sources, such as studies and documents that can support students' arguments or claims in their essays. They are used to search for information that can back up their statements with credible evidence and strengthen the overall validity of their essay.	Is there any science database to support 168 intermittent fasting?
		Are these data reliable?
		Is there any scientific evidence of the aromatherapy?

When analyzing prompts, the researchers coded them, based on both a thorough examination of the prompts themselves and the associated provided responses from ChatGPT that they generated. Subsequently, the researchers revisited the prompts with the labeled categories to ensure that the primary purposes of the prompts were accurately coded and assigned to the respective categories. The coding of the prompts was carried out independently by two researchers for each prompt, who later compared their results. In cases where discrepancies arose in categorizing the prompts, the researchers engaged in discussions to reach a consensus and resolve any differences in coding.

11.4 Results and Discussions

11.4.1 The Students' Interactions with ChatGPT: Analysis of the Prompts

The results of analyzing the prompts are presented in Table 11.2 and Fig. 11.6. As shown in the latter, when planning for the first writing task, participants predominantly employed prompts from the concept category (33%), followed by the standpoint (22%), argument (18%), and supporting detail categories (13%) were employed. On the other hand, when interacting with ChatGPT during the planning phase of their second writing task, participants most frequently employed prompts from the argument category (30%). In comparison, prompts from the concept and

Table 11.2 The percentage of each category in both writing tasks

	Aromatherapy		168 Intermittent Fasting	
	Number	Percentage	Number	Percentage
Concept	86	33.46%	35	27.78%
Standpoint	56	21.79%	7	5.56%
Argument	45	17.51%	38	30.16%
Supporting Detail	34	13.23%	36	28.57%
Rhetoric Sample	16	6.23%	5	3.97%
Evidence	20	7.78%	5	3.97%
Total	257	100%	126	100%

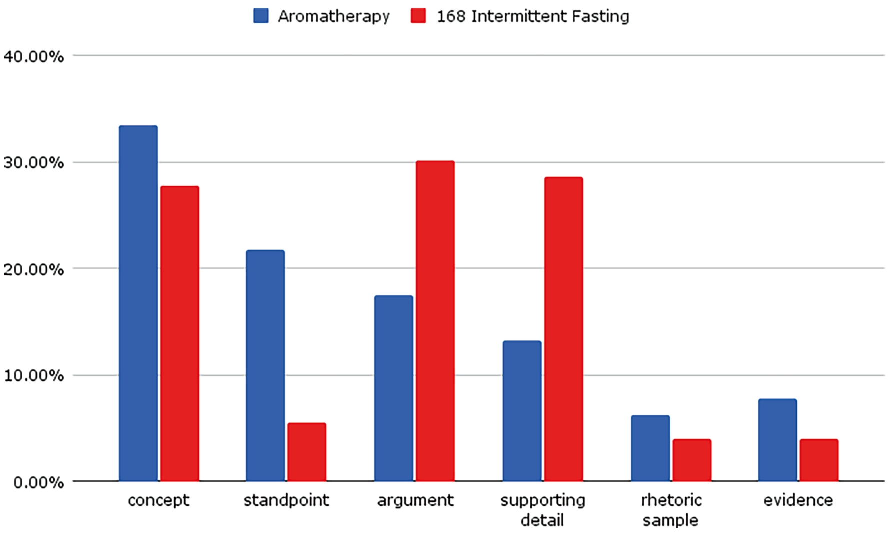

Fig. 11.6 The percentage of each category of prompt in both writing tasks

supporting detail categories are the second most commonly used (both approximately 28%).

Though differences exist in the prompt categories the participants generated in planning the two writing tasks, our work is descriptive: we are not evaluating, at this point, one category as superior to the other. Using prompts from different categories indicates distinct patterns or approaches to planning a writing task, but also individuals' experiences. For instance, if an individual has formulated a writing plan advocating for aromatherapy and presenting two arguments, they may pose more questions in the argument and supporting detail categories. On the other hand, if an individual is relatively unfamiliar with aromatherapy and opts to explore the topic before developing a writing plan, they may pose more questions in the concept and standpoint categories to gather more ideas to form a writing plan.

Still, a comparison of the prompts provided by the participants during their interactions with ChatGPT for the first and second writing tasks revealed a notable difference. During the planning of the second task, it was observed that participants entered more prompts toward building the argument and supporting detail categories. This emphasis on intentional argument-building suggests that the participants employed a more goal-oriented approach when planning the second writing task. This can imply that students have formed a writing plan before interacting with ChatGPT, leading them to ask questions aligned with their predetermined writing plans and goals. One plausible explanation for this result could be the explicit instruction on semantic mapping and the concepts of critical thinking and metacognition. With metacognitive awareness and guidance of semantic mapping, the participants could evaluate and make informed decisions on the information obtained from the interactions with ChatGPT.

Additionally, the analysis revealed that prompts in the concept category were used often in both writing tasks and were commonly provided at the outset of the planning phase when participants engaged with ChatGPT. This observation suggests that participants tended first to get acquainted with the topic. For instance, initial interactions with ChatGPT included questions such as 'What is 168 intermittent fasting?'. The responses obtained from ChatGPT could then be utilized in three ways: (1) providing foundational information to ensure students' comprehension of the method, mainly if they were unfamiliar with it; (2) employing the generated information as the introductory paragraph's definition of 168 intermittent fasting; and, (3) based on the obtained information, generating further prompts to engage deeper with the method by posing follow-up questions to ChatGPT.

Additionally, participants' prompts in the evidence category were limited in number. This observation suggests that participants recognized the limitations of ChatGPT, which does not provide sources for the information it generates, and therefore, responses may lack reliability. This finding is further supported by the questionnaire results, which revealed that the participants doubted the validity and reliability of ChatGPT's responses. As a result, they would evaluate the responses and seek additional information from other sources.

Moreover, despite the ability of ChatGPT to generate well-structured outlines and proficiently written paragraphs, the students rarely used prompts in the rhetorical sample category. We propose a couple of possible reasons for this observation. Firstly, the students were instructed to record all their interactions with ChatGPT and submit them to the instructors, which may have discouraged them from requesting ChatGPT to generate a sample outline, paragraph, or essay on their behalf. Secondly, the writing assignments clearly outlined and explained the task demands, and the students were already familiar with the rhetorical structure of expository writing due to their previous training and the instructions of this course.

11.4.2 *Participants' Perspectives of Their Engagement with ChatGPT: Questionnaire Analysis*

Regarding the function category, over 90% of the participants concurred that interacting with ChatGPT facilitated their information gathering and outline formation during the planning phase of their English writing tasks. Moreover, a substantial majority (88.5%) of the participants agreed that engaging with ChatGPT aided them in generating ideas for English compositions, and 84.6% of them acknowledged that the contents of their English compositions were strengthened due to interacting with ChatGPT. Regarding improving clarity and coherence, 65.4% of the participants agreed that interacting with ChatGPT improved the clarity and coherence of their English composition.

As for utilizing ChatGPT to search information sources and obtain examples for their arguments, nearly 70% (69.2%) of the participants agreed that the responses provided by ChatGPT helped generate examples. However, only a tiny percentage, 15.4%, thought that ChatGPT could assist them in accessing references, which is not included in the intended functionality of ChatGPT (Table 11.3).

In the engagement category, nine items explored the participants' experiences during their interactions with ChatGPT. Regarding the prompts they provided, approximately 80% of participants expressed disagreement with the notion of not knowing what to ask. Additionally, around 77% of participants reported that their questions were based on their writing plans, and 87.5% indicated that they would

Table 11.3 Items in the category of 'Function' and the results

Items	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree	Mean
1. Interacting with ChatGPT assists me in generating ideas when I plan for a writing task.	30.8	57.7	7.7	3.8		4.15
2. Interacting with ChatGPT assists me in forming an outline for my writing assignment.	26.9	65.4	7.7			4.19
5. Interacting with ChatGPT assists me in collecting information when I plan for a writing task.	34.6	61.5	3.8			4.31
6. Interacting with ChatGPT assists me in accessing references.		15.4	46.2	19.2	19.2	2.58
7. Interacting with ChatGPT assists me in locating examples to support my arguments.	15.4	53.8	23.1	7.7		3.77
11. Interacting with ChatGPT assists me in strengthening the content of my article.	11.5	73.1	15.4			3.96
12. Interacting with ChatGPT assists to improve the clarity and coherence of my writing.	7.7	57.7	30.8	3.8		3.69

Table 11.4 Items in the Engagement category and the results

Items	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree	Mean
3. I do not know what to ask when interacting with ChatGPT.			19.2	53.8	26.9	1.92
4. I have formed my writing plan before I interact with ChatGPT.	3.8	26.9	46.2	23.1		3.12
8. The questions I ask ChatGPT are based on my writing plans.	23.1	53.8	19.2	3.8		3.96
9. I ask further questions based on the responses provided by ChatGPT.	23.1	65.4	11.5			4.12
15. I do not know how to rephrase my questions when I am not satisfied with the responses by ChatGPT.		15.4	23.1	53.8	7.7	2.46
18. The responses generated by ChatGPT are too long for me to understand.		7.7	23.1	57.7	11.5	2.27
19. The responses generated by ChatGPT are irrelevant.		3.8	30.8	57.7	7.7	2.31
20. I find interacting with ChatGPT time-consuming.		15.4	26.9	50	7.7	2.50
22. I prefer to interact with ChatGPT in English when I am required to write an English article.	38.5	34.6	26.9			4.12

formulate subsequent prompts based on the responses ChatGPT has generated. It is worth noting that 61.5% of participants disagreed with not knowing how to rephrase their questions when they found themselves dissatisfied with ChatGPT's responses. In other words, approximately 40% of participants might experience difficulty rephrasing their prompts after not receiving the desired responses from ChatGPT.

Regarding the responses generated by ChatGPT, approximately two-thirds of participants (65.4%) disagreed with the idea that ChatGPT's responses were irrelevant. This suggests that most of the participants found ChatGPT's responses relevant and aligned with the prompts they posed. Regarding the ease of interacting with ChatGPT, around 70% of the participants disagreed that ChatGPT's responses were lengthy and challenging to comprehend. Moreover, 57.7% of the participants disagreed with the idea that interacting with ChatGPT was time-consuming. Furthermore, 73.1% of the participants preferred engaging with ChatGPT in English when planning their English writing assignments.

The results of the items in the engagement category suggest that a portion of the participants, around 30% to 40%, who were EFL learners, might face limitations in fully reaping the benefits of engaging with ChatGPT for planning their English compositions, potentially due to their limited English proficiency (Table 11.4).

The application category part of the questionnaire assessed participants' evaluation of their interaction with ChatGPT. Concerning the accuracy of ChatGPT's responses, 61.6% of the participants reported doubts about their accuracy. Additionally, 65.4% of participants disagree with relying on the information provided by ChatGPT without verifying its accuracy. The instructor emphasized the potential risks associated with placing complete trust in the responses generated by ChatGPT, and the results indicate that over 60% of the participants would evaluate the responses provided by ChatGPT.

Regarding the responses generated by ChatGPT, a significant majority (88.4%) of the participants agreed that they would assess the relevance of the responses to their writing plans. This indicates they could metacognitively monitor their engagement with ChatGPT by evaluating the responses. Moreover, 84.6% of the participants disagreed that they had trouble integrating the information provided by ChatGPT into their compositions. This indicated their proactive evaluation of the information's suitability and applicability for their writing purposes.

More importantly, a considerable majority (73.1%) of the participants expressed their satisfaction with their interactions with ChatGPT, and a majority (76.9%) of the participants agreed that they would continue to use ChatGPT during the planning phase when composing English compositions in the future. This substantial percentage indicates that the participants recognized the potential benefits of their interactions with ChatGPT in the planning process, affirming that ChatGPT can assist them in planning for their English compositions (Table 11.5).

Table 11.5 Items in the Application category and the results

Items	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree	Mean
10. I do not know (evaluate) whether the responses provided by ChatGPT are relevant to my questions.		3.8	15.4	65.4	15.4	2.08
13. I have difficulties incorporating the information generated by ChatGPT into my own writing.		3.8	11.5	73.1	11.5	2.08
14. I have doubts about the accuracy of the responses provided by ChatGPT.	23.1	38.5	26.9	11.5		3.73
16. I evaluate whether the responses provided by ChatGPT are relevant to my writing plans.	11.5	76.9	11.5			4.00
17. I use the information provided by ChatGPT without checking its accuracy.		11.5	23.1	46.2	19.2	2.27
21. I am satisfied with my experiences of interacting with ChatGPT in assisting me to plan for my writing assignment.	19.2	69.2	7.7	3.8		4.04
23. I will continue to use ChatGPT to assist me in planning for my English writing assignment after that.	34.6	42.3	23.1			4.12

11.4.3 Factors Impacting Prompt Generation

Prompts examination suggests that several significant factors influence the decision-making process of EFL writers. These factors include the goal and task demand, topic and content, rhetoric, structure and organization, and research and evidence.

These factors play a crucial role in students' writing planning. The goal and task demands guide students in determining the purpose and scope of their writing, helping them select appropriate content and structure their work accordingly. The choice of topic and content enables students to clearly understand the subject or participant matter, develop a focused central idea, identify standpoints or arguments, and effectively generate and organize their thoughts. Rhetoric structure and organization enable students to determine the overall structure of the writing. Also, it reminds the students to develop body paragraphs with supporting evidence or examples for overall coherence and clarity, allowing readers to follow the flow of ideas. Conducting research and selecting evidence help students bolster the credibility and potency of their arguments, showcasing their comprehension of the participant matter. These influential factors shape students' decisions, leading to well-planned, organized, and persuasive written pieces.

We contend that these factors are pivotal in shaping a writer's planning phase, influencing their interaction with ChatGPT as a writing tool, as exemplified in the sample prompts detailed in Table 11.6. These factors directly impact the prompts a writer selects when engaging with ChatGPT. For instance, a writer may seek assistance from ChatGPT in crafting a compelling outline for an expository essay or formulating an impactful introductory paragraph when the primary concern revolves around the category of rhetorical structure and organization.

By these factors and the delineated categories presented in Table 11.1, the subsequent section recommends prompts for writers to utilize when seeking interaction with ChatGPT to aid in planning their writing tasks.

11.4.4 Suggested Prompts and Models for Engaging Learners with ChatGPT in the Planning Phase of Writing

Based on the findings of this study, we propose a model for utilizing ChatGPT in the planning phase of writing (Fig. 11.7). The model has four components: ChatGPT as a tool, a monitoring mechanism consisting of critical thinking and metacognitive monitoring, a factor-controlled generator of prompts, and the planning process.

Table 11.6 Factors that impact the decision making in generating prompts

Factors	Descriptions	Samples
Goal and Task demand	To guide students in determining the purpose and scope of their writing, helping them select appropriate content and structure their work accordingly.	If I want to write an article to persuade people who deeply believe in the effectiveness of aromatherapy, to stop doing it, what should I do?
Topic and Content	To define the topic or participant matter and develop a clear, focused central idea.	What is the 168 intermittent fasting?
	To identify standpoints or arguments	What are the benefits of 168 intermittent fasting? Why can 168 intermittent fasting induce autophagy?
	To generate and arrange the thoughts	Is 168 intermittent fasting suitable for everyone? Why are people skeptical about aromatherapy?
Rhetoric Structure and Organization	To determine the overall structure of your writing.	Give me the 4-paragraph essay explaining why we can adopt 168 intermittent fasting.
	To develop body paragraphs with supporting evidence or examples to ensure coherence and clarity, allowing readers to follow the flow of ideas.	The evidence and research about 168 intermittent fasting. 168 intermittent fasting measure more details about appetite control.
Research and Evidence	To gather relevant information (e.g., evidence)	Is there any study suggesting that 168 diet could help our body?
Others	Personal interest	What is the best time to do it? If I use intermittent fasting, will I faint during exercise?

In this model, ChatGPT serves as the tool that aids a writer in planning a writing task. A planning process involves three subprocesses: generating ideas, organizing ideas, and goal setting. When a writer engages with ChatGPT during each subprocess, the factors influencing writing (depicted by the blue ovals in the model) can shape the writer's selection of prompts for interaction with ChatGPT. Hence, the writer would generate prompts corresponding to various categories (aligned with the factors of writing in the model) to assist in planning the current writing task. Additionally, in this model, we underscore the significance of cognitive monitoring, encompassing critical thinking and metacognitive awareness. A writer's cognitive monitoring supervises their writing plans and the factors of writing, determining

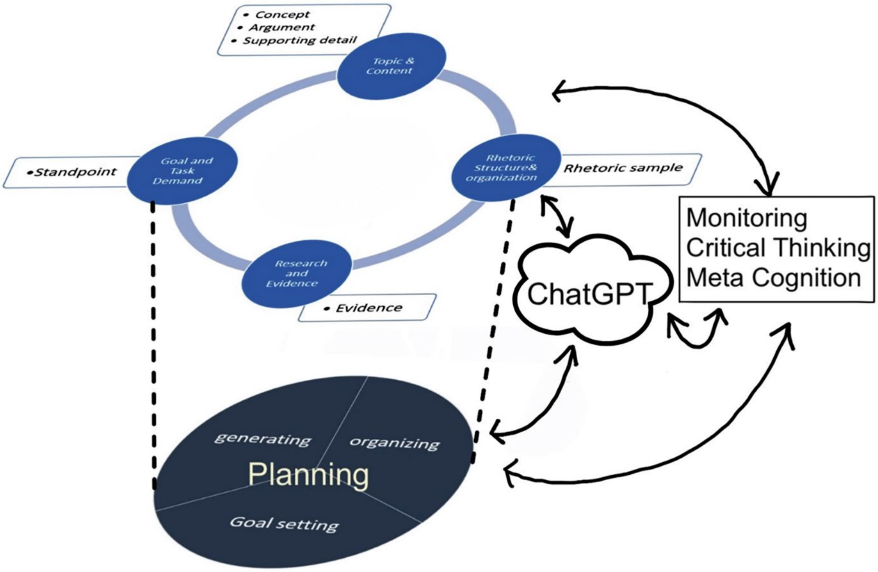

Fig. 11.7 The writing model with ChatGPT in the planning phase. © 2024 Yu-ling You, Axson Lin, Shu-Jhen Han, Chiao-Chu Chu, and Chih-Yun Dai

whether the prompts generated by ChatGPT align with the task demands and the writer's goals and plans.

Through the presentation of this model, our aim is to underscore ChatGPT's capacity to serve as a valuable aid during the planning phase. We emphasize that a writer's cognitive monitoring is indispensable when integrating ChatGPT into the writing process. Armed with critical thinking and metacognitive awareness, individuals can make informed decisions regarding the responses generated by ChatGPT, aligning them with the pertinent factors that need consideration during the planning process.

Based on the prompts formulated by the participants during their engagement with ChatGPT, we have identified several prompt patterns that learners can employ to interact with ChatGPT when planning their writing tasks. Prompts play a crucial role in enabling users to effectively use ChatGPT as a tool to obtain the necessary information. The suggested prompts listed in Table 11.7 are designed to assist learners in effectively requesting ChatGPT to search and gather relevant information, thereby facilitating the planning phase of their writing tasks.

The suggested prompts are not exhaustive; rather, they can serve as a starting point for learners when using ChatGPT as a tool to assist them in the planning phase of composing. The prompts are categorized based on key factors relevant to composing a written piece. This allows learners to selectively choose prompts from categories aligned with their specific concerns, seeking suggestions with the

Table 11.7 Suggested prompts in the first writing task

Prompts	Syntactic Structures	Examples
Concept	(1) What is [concept] ...? (2) Define [concept] ...	(1) **What is** aromatherapy? (2) **Define** aromatherapy. (2) **What is** 168 intermittent fasting? **(3) What is** lean body mass?
Standpoint	(1) Do you think [standpoint].......? (2) Can a concept or an idea [standpoint] ...? (3) Is a concept or an idea [standpoint]?	(1) **Do you think** aromatherapy can improve sleep quality? (2) **Can** aromatherapy help with relieving stress or pain? (3) Is 168 intermittent fasting **useful**?
Argument	(1) What is/are [arguments] of a concept or an idea? (2) The [argument] of a concept or an idea.	(1)**What are the benefits of a**romatherapy? (2) What are **the downside of** aromatherapy? (3) **What are the advantages of** implementing 168 intermittent fasting? (4) **What are the disadvantages of** conducting 168 intermittent fasting?
Supporting Detail	How/Why do..... support/reject a concept or an idea? How can a concept or an idea [supporting detail]?	(1) **How do** essential oils **support** immune function? (2) **Why do** people **reject** to use aromatherapy? (3) **Why is** intermittent fasting **efficient** in losing weight? (4) **How can** 168 intermittent fasting **enhance health indicators**?
Rhetoric Sample	Imperative sentence: write ... [rhetoric sample].	(1) **Write a five-paragraph essay on** aromatherapy. (2) **Provide an outline** for an argumentative essay on aromatherapy.
Evidence	What is the [proof/data/evidence of] ....? Imperative sentence: Give/ Provide ... [rhetoric sample].	(1) **What is the evidence** to support the benefits of aromatherapy? (2) **What is the scientific evidence** that aromatherapy can help to relieve pressure ? (3) **What are the proofs** of applying 168 intermittent fasting for a long period? (4) Give me **evidence** for each of the arguments that support 168 intermittent fasting.

assistance of ChatGPT. For learners with limited experience using ChatGPT for writing task planning, this list of suggested prompts can guide them in effectively obtaining the information they need from ChatGPT.

11.5 Conclusion

This study explored the utilization of ChatGPT during the planning phase when EFL learners plan for their English compositions. Through analysis of the prompts generated by participants during their interactions with ChatGPT, we categorize and propose prompts that can be employed to request an inquiry from the chatbot. Furthermore, we examine the factors influencing the types of inquiries students pose. Finally, based on the prompts and factors, we propose a model that emphasizes the role of metacognitive monitoring when language instructors and learners attempt to integrate ChatGPT into the planning phase of EFL learners' English composition writing.

The results of the questionnaire, which investigated learners' perceptions and feedback regarding their engagement with ChatGPT, provide substantial evidence supporting our assumption that ChatGPT is a valuable tool in assisting EFL learners to plan for their English writing tasks. Firstly, ChatGPT serves as a proficient language model, providing proper English input, and functions as a language resource, offering an extensive collection of English material for EFL learners. Secondly, ChatGPT assists EFL learners in making individualized writing plans that align with their specific interests and the writing goals they set. This personalized support is possible as ChatGPT can offer tailored assistance and suggestions based on the prompts provided by individual learners. Thirdly, ChatGPT supports EFL writers' planning in exploring potential ideas and organizing their writing ideas, for instance, by providing a semantic map. Lastly, ChatGPT responds to their rhetorical demands, presenting outlines, sample paragraphs, sample essays, etc., for learners to observe and imitate the rhetorical structures and styles used in writing.

The model of incorporating ChatGPT into the writers' planning phase of composing highlights the significance of monitoring, which primarily encompasses critical thinking and metacognitive awareness. The efficacy of obtaining intended and useful information from ChatGPT depends on learners' prompts. Consequently, to effectively utilize ChatGPT in the planning of their writing tasks, learners need to analyze, evaluate, and synthesize the information generated by ChatGPT, which requires them to think independently, reason logically, and make informed decisions. Moreover, this monitoring process requires an individual's conscious awareness, enabling them to employ critical thinking effectively. This cognitive thinking and metacognitive monitoring distinguish humans from machines and is the key for educators to effectively harness AI, such as ChatGPT, in teaching and learning.

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